

Re-Designing A Mobile App with Patients with Discordant Chronic Comorbidities: Usability Study

Tianshuo Li*
lit017@udayton.edu
University of Dayton
Dayton, Ohio, USA

Mason Haynes
University of Dayton
Dayton, Ohio, USA

Bradley Rucker
University of Dayton
Dayton, Ohio, USA

ABSTRACT

Patients with complex conditions and treatment plans often find it challenging to communicate with multiple providers and to prioritize various management tasks. The challenge is even greater for patients with discordant chronic comorbidities (DCCs), a situation where a patient has conditions that have unrelated and/or conflicting treatment plans. We identified possible needs and designed the Apps that addressed those needs (including: goal setting, ease of use, monitoring, and motivation). We then tested this App with patients with DCCs to see if those needs were addressed. We present results from that a usability study that highlight the design preference of patients with DCCs.

CCS CONCEPTS

• **Human-Centered Computing** → Human computer interaction (HCI) .

KEYWORDS

Mobile Application; Discordant Chronic Comorbidities, Patients Care and Support, Multiple Chronic Conditions, User ability study;

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1 INTRODUCTION

A major challenge in society as of now deals with discordant chronic comorbidities (DCCs). A DCC can be described as a health condition in which a patient has two or more, usually unrelated chronic diseases/conditions that must be treated simultaneously and also have potentially conflicting treatments between the patient's illnesses [3, 4]. For example, Type-2 diabetes, Arthritis and Depression are discordant to each other. Patients may work on getting their diabetes symptoms under control with one provider, while simultaneously working with another provider on controlling their arthritis. As such, any changes in the treatment plan for diabetes often negatively impacts the control of their arthritis. In attempts

to combat this problem, we designed and evaluated a mobile application to address the DCCs care issues by specifically drawing the attention of professional support networks to the needs of multiple providers supporting the care and treatment of patients with DCCs. The study on the patients entailed an interview pertaining to the usability of the app. Their responses were reviewed and concerns are addressed later in this paper.

2 BACKGROUND

The majority of Americans have a mobile device of some kind making it a viable platform for such a tool. This team began researching to see what has already been found in studies. A 2018 study by Ongwere et al.[4] found patients struggle to follow doctors orders when they receive competing information. In this study one patient with the DCC combination of being diagnosed with both Type-2 Diabetes, being directed to eat a lot of spinach, and end-stage renal disease, being directed by a separate provider to limit his intake of spinach.[4] This study further found evidence of little to no communication between providers.[4] Another study found this same issue where patients found different providers had little to no coordination between them.[5] Such lack of communication leads to potential complications for patients. Further, “patients with DCCs struggled to prioritize and balance often-conflicting treatment plans...”[5]. The failure to find the correct balance often results in lower quality of life as noted in the same study drug interactions have the potential for causing worsening of symptoms in their current diseases or worse present their new severe symptoms.[5] Studies have pointed to mobile apps including the ability for patients to track medications they are taking currently as well as historically, and for these apps to have the ability to display the information quickly to help physicians gathering medical history.[1] Such apps would also provide feedback, and tracking to help patients with self-monitoring.[2] We identified possible needs and designed the Apps that addressed those needs. For example, monitor multiple DCCs simultaneously, alert patients when possible prescriptions or medical instructions conflict and expedite the resolution of such negative interactions. We then tested this App with patients with DCCs to see if those needs are addressed.

3 METHODS

In this report, we present an extension and implementation of the work done by Ongwere et, al.[4, 5]. Ongwere et, al. addressed the challenges of supporting the care and management of patients with DCCs. We have created and evaluated a mobile app, focused on addressing challenges pointed in those studies, we utilized their design suggestions for areas that need to be improved to tailor our design for the care and management of DCCs. We then conducted

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an interview with patients with DCCs (N = 12) to evaluate the function use and functionality of the app. After showing them the App we asked the participants to show how they would use them to address their needs, we asked them to vocalize their thoughts as they use them. We observed as the participant demonstrated how they would use the Apps and asked the following question?

- RQ1: How can you use this design to set goals and navigate this situation during the time of crisis, during times of change, and when symptoms are stable?
 - RQ 1.1: What are the steps for setting and implementing these goals?
 - RQ 1.2: What are the potential constraints to following these steps?
 - RQ 1.3: What needs to be done to get a patient to do those steps?
 - RQ 1.4: If a patient could make changes to this prototype, what would they change?
- RQ2: How can Jeff use this design to see across their conditions that may potentially conflict and then balance or make decisions based on those conflicts?
- RQ3: If implemented, will his design help Jeff prioritize his treatment and communicate with their multiple healthcare providers?
 - RQ3.1 How likely is it that you would want to use this prototype?
 - RQ3.2 If you were to use a tool like this, how might it help prioritize your treatment and communicate with your multiple providers?

The questions outlined below were devised to get comments about all of the different uses and sections of the app. They were created to allow for people to envision themselves using the app regularly and seeing what would work well and what wouldn't work as well. The study was approved by the University IRB.

4 FINDINGS

The following themes or topics emerged from analyzing the audio transcripts from the patient interviews, i) user interface and aesthetics, ii) prioritization and goal setting, iii) functionality and motivation, and iv) APIs.

4.1 User Interface and Aesthetics

4.1.1 Simplicity and easy to use. One of the major compliments that the app has received is in regards to the applications simplicity and ease of access. The basis of this complement is as follows,

“Because the simpler this app has the implementation is kind of like a calendar that reminds me of my day to day activities. So I can, I can just like navigate my treatment process through this section and combine it with other functionalities of this app is great.” - P2

Another concern presented by a participant when commenting on the functionality of the app in general was as follows:

“I think, I think I kind of touched on that a little bit. Making it as simple as possible is probably the best.” -P12

One of the concern that we saw in our study, that participant did not know exactly what the app, for explore one participant explained that:

“As a person who knows nothing about the application, before downloading the application, I need to understand the center and value of the application for the first time.” - P7

On the same topic of the last quote, another participant comments on issues with the ease of use of the app, by saying:

“I feel like it is too technologically savvy meaning that people will have to have a lot of training to use the app. That's why I think it can be made easier for people who aren't savvy to use it easily.” -P1

4.1.2 Automation. Automation is a big issue in making an app and attempting to make it user friendly at the same time.

On the positive there were some negatives and complaints raised by some participants. One participants commented on the issue of automation by saying,

“...the issue with the fact people have to write so much everyday to enter their information. It may be advantageous to make the majority preselects with the option for expounding on the information . It would also help us with data analysis standardizing input for machine learning models, etc.” - P3

4.1.3 Look and feel. The look and feel of an app are what keep people interested and are what make people attracted to using an app. Overall, there were some comments from the participants about the look and feel of the app. There were some good and some bad comments about this topic. One compliment about the look and feel of the app is as follows:

“In that sense, but otherwise, I think it looks nice. In general, and the way you guys presented it looks really nice. But I think, I think it's great.”- P10

With this comment came additional complaints and suggestions in regard to the look and feel of the application. One such comment in particular pertains to the look and feel of the design is as follows,

“I'm just recommending you change the font, because here it looks like it's overlapping itself and then people will say, Who are colorblind or have bad eyesight may not be able to see that correctly. They may be confused on what it's saying because it could be blurry to them”- P8

Additionally, other participants saw an issue in regards to the color tones and color combinations on the various menus and screens of the app. They said,

“Again with the color scheme. Try not to like to keep to this one color and especially like two darks or two lights and then everything in between. I recommend finding something vibrant because you also want to keep in mind that even though I know you're trying to have that low soft cool mood of a color”- P8

Lastly, one additional area of the look and feel of the app that was commented on is the use or misuse of the white-spaces on the various menus in the app. The comment is as follows,

"Okay, Now Again, it's just cool this is good. However, there's a lot of white space. Again, white space. So what you want to do is you want to utilize some space all this space right."-P8

4.2 Prioritization and goal setting

4.2.1 Goal setting strategies. In the category of prioritization and goal setting, the first concern and compliment that was addressed is the fact of how the app is both good and bad in providing a strategy for setting goals on the app. One compliment of the state of the goal setting strategy that the app currently presents is as follows,

"So whenever I have problems, I can check those priorities and then to see what exactly I should do so it will just kind of categorize and display your problems layer by layer. So when you finish the first layer, you can do the second. So I think that's good" - P2

Along with some compliments came some problems with the goal setting strategy that the app provides. One complaint pertains to the person ability of the app in regards to setting goals. One participant comments on this issue by saying,

"See, I mean they're feeling good. I'm feeling bad. And then the next one said, Why am I feeling good?" Why am I feeling bad? And then from there, the doctor, whoever in the physician or clinical aspect of it. Can say okay, now I can make my assessment on you know, why is this person's feeling good, bad or not feeling nothing at all just none."- P8

In addition to the comment above relating to the person-ability of the app, other participants commented that the app doesn't provide any ways to accomplish your goals besides trial and error. One participant says that,

"...they can just add more professional courses. Or like health tips or like videos that contain health knowledge that teach the patients every day. Like, you can put 10 five minute short videos in that and teach the patient's of how they can remain healthy and accomplish their goals."-P2

4.2.2 Monitoring. Another facet of the comments related to prioritization and goal setting is in relation to the monitoring and tracking of symptoms or activities. Overall, there were compliments and complaints about the current state of the monitoring aspect of the app itself. One participant compliments the monitoring aspect of the app by saying,

"I think I can use this app to monitor my symptoms and I'll be able to notice over time. If there's a change in anything that's going on. And then I'll be able to contact the doctor or figure out how I can, I guess fix what's going on so he can be better off." - P3

One major complaint in regards to the monitoring aspect of the app pertains to the fact that the app doesn't go into much detail about why you are experiencing these symptoms and what specifically the symptoms are and how they make you feel. One participant comments on this by saying,

"So you can say non happier or good or bad, like non good or bad or, you know, something like that. And then then as you go into the second conditions or prior to that, then you have them explain what does the non mean what does the good mean, where does the bad mean right." - P8

Additionally, participants were connecting on the fact that the app monitoring is not as streamline as it could be. One participant talks about this by saying that,

"For example, it will not be just Like, like your, your information, blah, blah, blah, blah. But it'll be like what you did yesterday. What you do today, what you should do tomorrow. So that it's like a streamline. It's like a timeline." - P2

4.3 Functionality and Motivation

4.3.1 User motivation . Motivation is a difficult thing to feel especially when it pertains to using something everyday. One of the most commented aspects of the app is in regards to motivation and getting the user to be motivated to use the app on a daily basis. One participant comments on the existing source of motivation as promoting one's own health and well being by saying,

"I think I can use this app to monitor my symptoms and I'll be able to notice over time. If there's a change in anything that's going on. And then I'll be able to contact my doctor or figure out how he can fix what's going on so I can be better off."

One common addition to the existing motivation that the app possesses, as described in the quote above, an additional source of motivation could come in a reward system. One participant touches on this point by saying,

"You'd like an incentive to do things like if there was like an incentive system to to like push people to do more than they will employee obviously like being healthy and like, that's a big incentive if there's like something little wake up call answer something every time you do it and you can personalize your app different Like it's like to actually be another way to ask the constraints of not being motivated to actually go and lace up your goals." - P4

In addition to this quote another participant comments on adding a reward system by saying,

"... A little bit like the incentive system. I feel like health-care professionals were pushing harder and be like, you should be doing this on it like actually." - P4

One last comment on improving the motivation generated by the app relates to reminders and alarms to remind the user to do some activities that will help them to have good days and avoid bad days. One participant comments on this issue by saying,

"How long I need to exercise every day to keep healthy and then it has to be implemented, which will remind me of the upcoming events I need to do. So it just kind of pushes me And reminds me of every detail so that I can break these constraints and to constantly fight to improve my situation." - P2

4.4 Using existing APIs

4.4.1 Communicating with multiple providers. Integration with existing systems can help users obtain information faster and more conveniently. One participant concern about the connection between application and health provider system,

"I think one of the biggest challenges is the doctor side of the application because typically doctors and medical facilities have their own system."-P12

That participant also used the EPIC system from Johns Hopkins Medicine as an example to explain how other systems solve this problem.

"And epic is kind of doing a similar thing where they're trying to connect databases of different Medical Facilities. But they have a kind of cross-platform mentality where they will connect you know whatever platform. The doctor is using the platform that or it can potentially connect to whatever platform."-P12

One last facet of the prioritization and goal setting ability of the app pertains to the fact that there is not the ability to connect your account to multiple providers, in the event that the user needs to. One participant talks to this issue by saying,

"The information section, they can have a link to the doctors phones. So there the doctor can also download this app, and then it can monitor his patients health condition every day. So it will be like the app for patients will not be isolated because it can. If you add this function in your information section you can link yourself with your doctor."- P2

To further the point above one participant brings up the issue of not being able to connect with insurance and hospitals directly. That participant say the following,

"If possible, connect with insurance information(all in one)." - P6

5 DISCUSSION

5.1 User Interface and Aesthetic

The user interface and the aesthetics of the app design are an aspect that will undergo the most changes and alteration. The items that the participants found appealing for user friendly interfaces include: i.) Simplicity and Ease to Use, ii.) Automation, iii.) Look and Feel iv.) Buttons v.) Prioritization and Goal Setting, and vi.) Using existing APIs.

5.2 Simplicity and Ease to Use

Overall, simplicity is on the forefront of our vision in regards to generalizing the app and giving it the ability to be used by every individual. One of the major compliments of the app was that the app is structured to be used every day so it puts the user into a routine of using the app and inputting data and such, said by Participant 2. This compliment is the basis that we will build upon to make the app even simpler. While using some of the criticism given by our participants we saw that a common connection is that participants found that the user will need an introduction to the app and how to use it before they begin using the app on a regular basis,

comments also included utilizing the white space on the menus. This complaint can be seen in the following quote by Participant 7 seen in section 4.1.1. Our solution to this and an alteration we look to implement into the app is the use of hints or suggestions to the user. One possible solution to these problems is to add a character to the app that will be seen as the guide as seen in. Having helpful and interactive hints will help the user to not feel like it is them who are not smart enough to use the app. Overall, this solution will address the participants' concerns addressed above and will allow for the white space on the menus to be utilized properly.

5.3 Automation

In regards to the app, automating the app was brought up by Participant 3 in a quote referenced in Section 4.1.2 One solution to this problem could come in the form of a new shortcut menu. This menu would house custom, user created shortcuts that they would set up and would allow for the user to click on that shortcut and it would automatically fill that into the app. For example, if a user would go for a 30 minute run every morning at 8 am they could set a preset or shortcut that they would create and just tap on after their run and it would be added to the activities section of the app.

Another solution to this problem is linking the popup keyboard to a voice keyboard. This would allow for the user to speak to the app and it will take their words and type for them. This would further increase the ease of use of the app as well as make the app more automated by allowing for the user to just speak instead of type everything out. The implementation of these solutions would answer the criticism of the participants and would make the app be more autonomous.

5.4 Look and Feel

The last aspect of the app that will be changed is the general look and feel of the app. Overall, our participants said that the app looks good but there are places that need to be cleaned up to make the app look more professional and be ready for mass distribution. One example of this comes in a quote by participant 8 in Section 4.1.3. In this menu it can be seen that the boxes are not all the same with the text in them. Some are centered justified while others are not. Additionally, All of the words are not on the same level in each box. This is just one example of a menu that needs to be cleaned up to allow for a more professional look and feel. Using all of the input from the participants all of the menus of the app will be looked at to ensure a professional look and to ensure that there is continuity amongst all of the menus.

5.5 Buttons

The last change that will be made to the user interfaces and the aesthetics pertains to the buttons across the app. The participants brought up the fact that the buttons on various menus of the app are difficult to see and locate. The following menu is an example brought up by Participant 8 in Section 4.1.3.

The main solution to this problem is to replace the existing buttons with ones that have more contrast and distinction. This will allow the buttons in the app are to have more contrast and be used to take up some of the white space that can be seen on certain menus, as referenced in Section i. These buttons will pop off the

screen and have contrast to help people with vision problems to see them and to be able to obviously discern them from the rest of the words on the screen.

5.6 Prioritization and Goal Setting

During development of our application this was a defining backbone of our design. We focused on our users being able to sufficiently set, track and prioritize their goals with help from the app such as reminders of daily medications, workout schedules and food intake. In our study we derived from participant feedback that our approach to goal setting as well as tracking strategies is in need of improvements. Participants felt that adding variety into the way in which we inform and teach users about living healthier lifestyles would be very desirable feature set. We plan to better inform the end user, by adding short videos on various health strategies such as dieting, exercise, mental health and regiment making. Participants further felt we could better represent the information we collect on them in a more streamlined and digest-able fashion. We plan to allow physicians to view patient information on a regular basis to make better assessment of the state of their patients conditions. We also plan to send updates when a possible trend life threatening behavior make changes to the patients homeostasis. These features will help both patient and physician communication and the continuity of healthcare in and outside of the doctor visits.

5.7 Using existing APIs

According to the suggestion and issue we got from participants, we will spend time on connecting with different healthcare providers and insurance information sessions. Regarding the connection with different healthcare providers, we have a long-term and detailed plan. In the next few months, we will cooperate with some local healthcare providers in Ohio and use the APIs they provide to build connections. At the same time, in these few months, we will add some functions to our application. For example, we will use OCR to import prescription and send generated pdf reports to the doctor. Those functions will temporarily instead of the connection with the different healthcare providers. Currently, the insurance information session is in progress, now UI design has been completed. For this function, users can fill in the relevant information such as the name of the insurance company, the website of the insurance company, and upload the insurance card. After adding those information, users can view insurance information and get notified when insurance expires.

6 CONCLUSION

Overall, the usability of our app was looked at in this study. We interviewed 12 patients about the usability of the app and any problems that could arise from using the app. The most important areas that participants focused on are: User Interface and Aesthetics, Prioritization and Goal Setting, and Functionality and Motivation. All of the responses pertaining to these categories were analyzed and some common resolutions were found to respond to the patients concerns. The resolutions are outlined in the discussion section and will be implemented into another design of the app.

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